

Requirements Specification

Webhook Ingestion & Retry Mechanism Hub

STUDENT	SRN	SECTION
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Functional & Non-Functional Requirements

FR-001	Register Webhook Integration	High	Functional
DESCRIPTION	The system shall allow an Integration Developer to register a webhook integration by specifying its source identifier and destination service endpoint.	ACCEPTANCE CRITERIA	A valid integration can be created and is available for routing. An invalid or missing destination endpoint is rejected.
RATIONALE	Provides the routing configuration required by the webhook gateway.		
FR-002	Ingest Webhook Event	High	Functional
DESCRIPTION	The system shall accept incoming webhook events and create a persistent delivery record containing the payload, received timestamp, integration identifier, and current delivery status.	ACCEPTANCE CRITERIA	Every accepted webhook produces exactly one stored record containing all required fields.
RATIONALE	Ensures each received event can be traced through processing.		
FR-003	Deliver Webhook	High	Functional
DESCRIPTION	The system shall route each accepted webhook event to the destination endpoint configured for its integration and record the outcome of the delivery attempt.	ACCEPTANCE CRITERIA	The configured target receives the original payload and the delivery result is recorded.
RATIONALE	Provides the core delivery function of the gateway.		

Requirements Specification — continued

FR-004	Retry Failed Delivery	High	Functional
DESCRIPTION	The system shall automatically reschedule a failed webhook delivery using progressively increasing retry delays and record every retry attempt.	ACCEPTANCE CRITERIA	A failed delivery produces further attempts with increasing delays, and each attempt is recorded.
RATIONALE	Allows temporary target-service failures to recover without losing events.		
FR-005	Manage Dead-Letter Queue	High	Functional
DESCRIPTION	The system shall move events that exhaust all permitted retry attempts into a dead-letter queue and allow a System Operator to inspect and manually replay them.	ACCEPTANCE CRITERIA	An exhausted event appears in the dead-letter queue with its failure history, and an operator can initiate replay.
RATIONALE	Provides controlled recovery for events that cannot be delivered automatically.		
NFR-001	Ingestion Capacity	High	Non-Functional — Performance & Reliability
DESCRIPTION	The system shall sustain an ingestion load of 2,500 webhook requests per second without losing accepted events.	ACCEPTANCE CRITERIA	During a load test at 2,500 requests per second, every accepted webhook is represented by a stored delivery record.
RATIONALE	Supports dependable operation under high-volume webhook traffic.		
NFR-002	Role-Based Access	High	Non-Functional — Security
DESCRIPTION	The system shall restrict access to webhook configuration, payload records, retry history, and dead-letter queue operations to authorized users according to their role.	ACCEPTANCE CRITERIA	Integration Developers and System Operators can perform only permitted operations; unauthorized attempts are rejected.
RATIONALE	Protects webhook data and operational controls from unauthorized access.		